

Skills Review

Exercise No. 7

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Grade:

Course/Section: CPE4B

Date: Sept. 12, 2026

Machine Problem:

Create a program that will display the output format shown below. Save the program using Exer7.asm as the filename.

```
Az
  By
    Cx
      Dw
        Ev
          Fu
            .
              .
                .
                  Za
```

Program:

```
; AARON LUDWIG A. ALTAR
; CPE4B

.model small
.stack
.code

start:
    mov ah, 02h
    mov cx, 26      ; 26 letters, A to Z

    mov bl, 7ah     ; bl = 'z'
    mov dl, 41h     ; dl = 'A'
    mov dh, dl      ; dh = 'A'

back:
    int 21h         ; display uppercase (dl)
    mov dl, bl      ; load lowercase
    int 21h         ; display lowercase

    dec bl          ; next uppercase letter
    inc dh          ; next lowercase letter
```

```
    mov dl, 0ah
    int 21h

    mov dl, dh
    loop back

    mov dl, 0dh
    int 21h
    mov dl, 0ah
    int 21h

    mov ah, 4ch
    int 21h

end start
```

Output:

```
Dw
Ev
Fu
Gt
Hs
Ir
Jq
Kp
Lo
Mn
Nm
Ol
Pk
Qj
Ri
Sh
Tg
Uf
Ve
Wd
Xc
Yb
Za
```